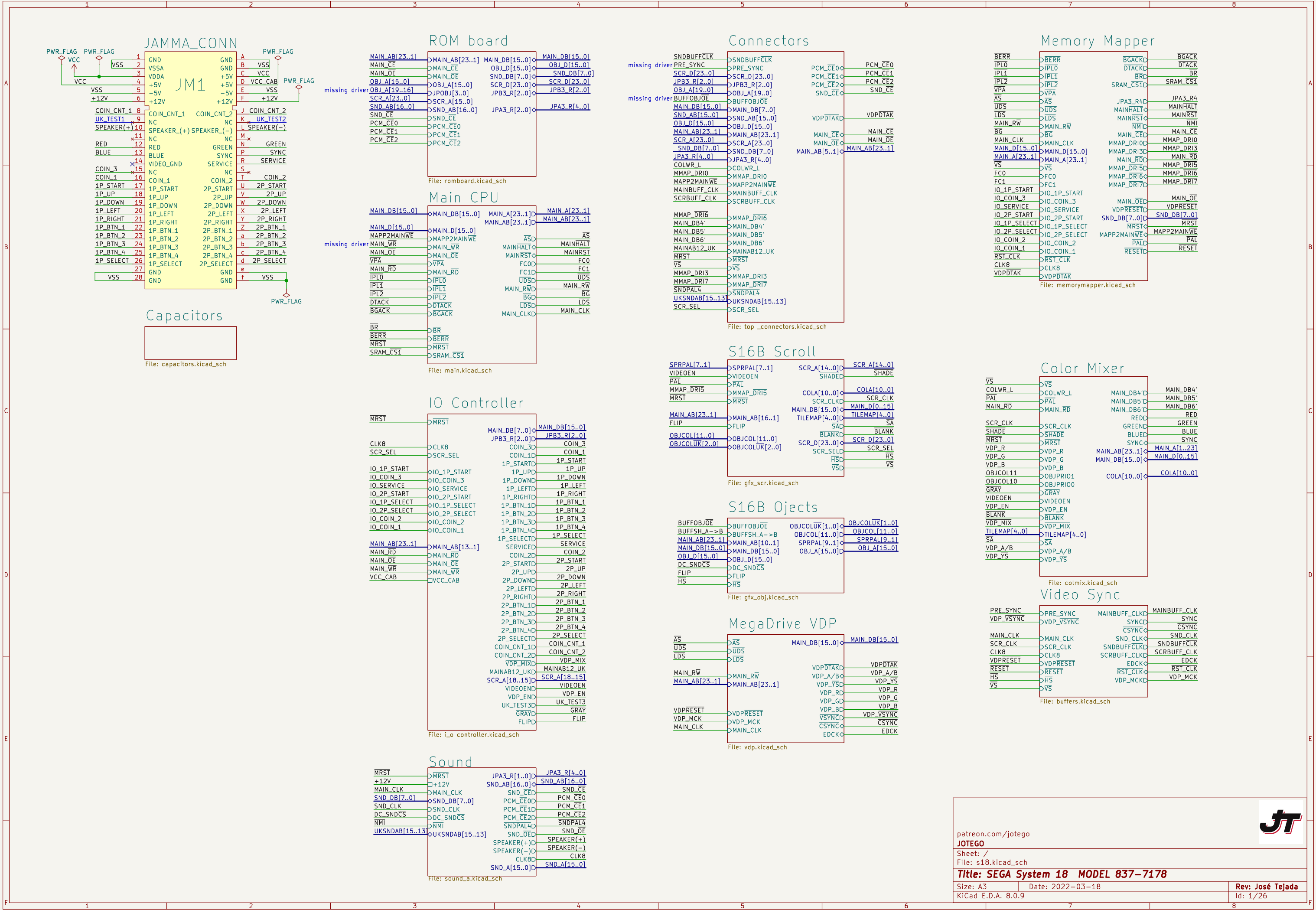
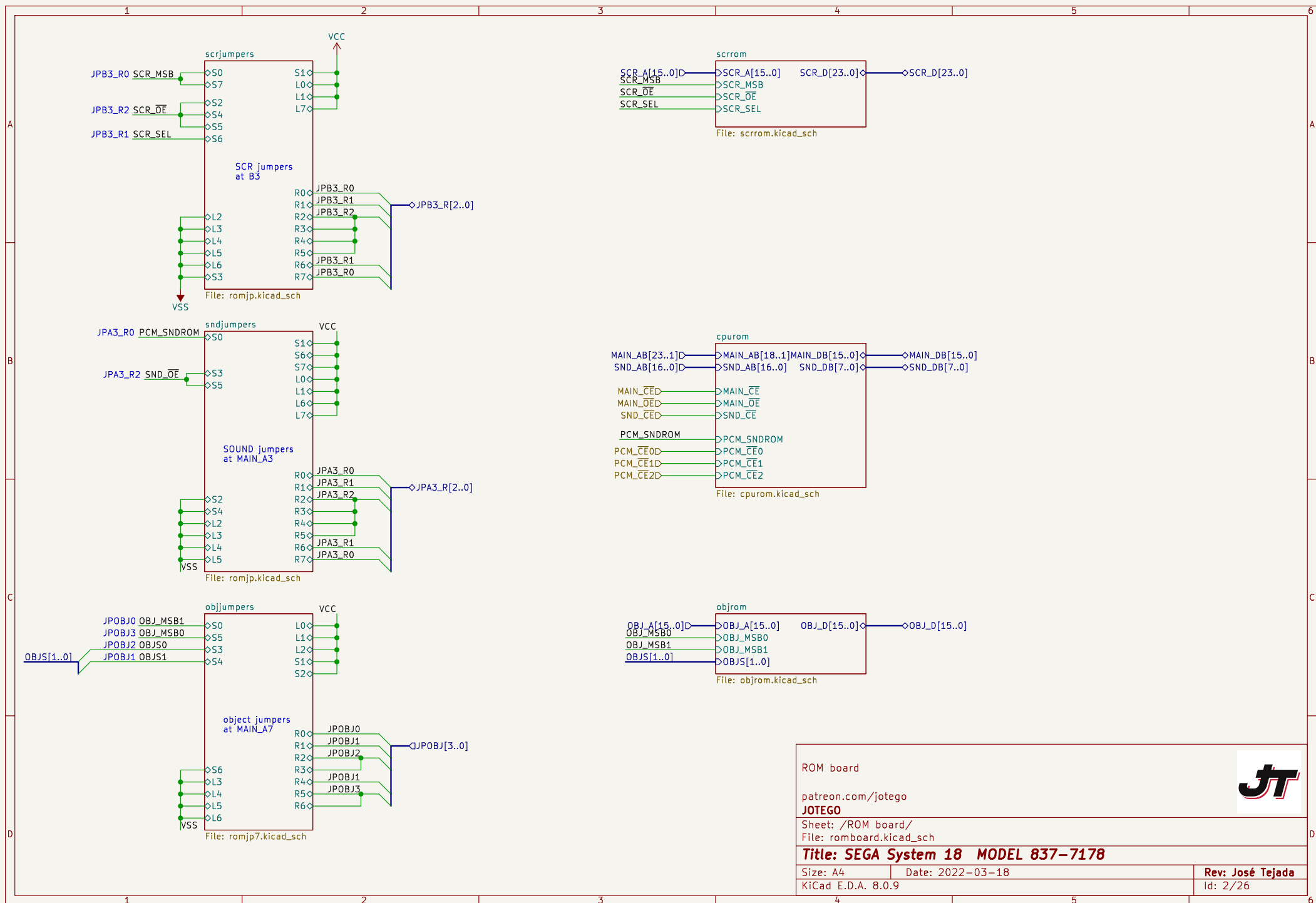






ROM board

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Sheet: /ROM board/

File: romboard.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4

Date: 2022-03-18

Rev: José Tejada

KiCad E.D.A. 8.0.9

Id: 2/26





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Sheet: /ROM board/scrjumpers/

File: romjp.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4

Date: 2022-03-18

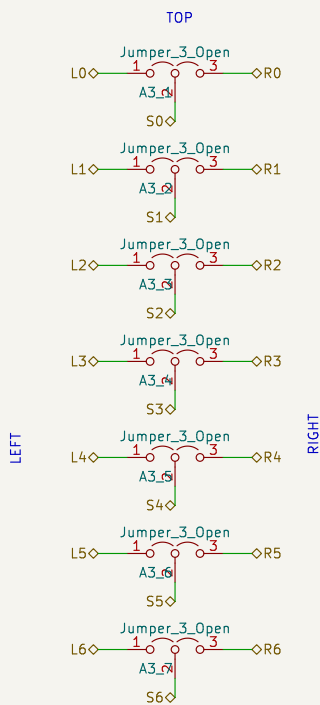
Rev: José Tejada

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Id: 3/26







BOTTOM

Seven jumpers

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Sheet: /ROM board/objjumpers/
File: romjp7.kicad_sch

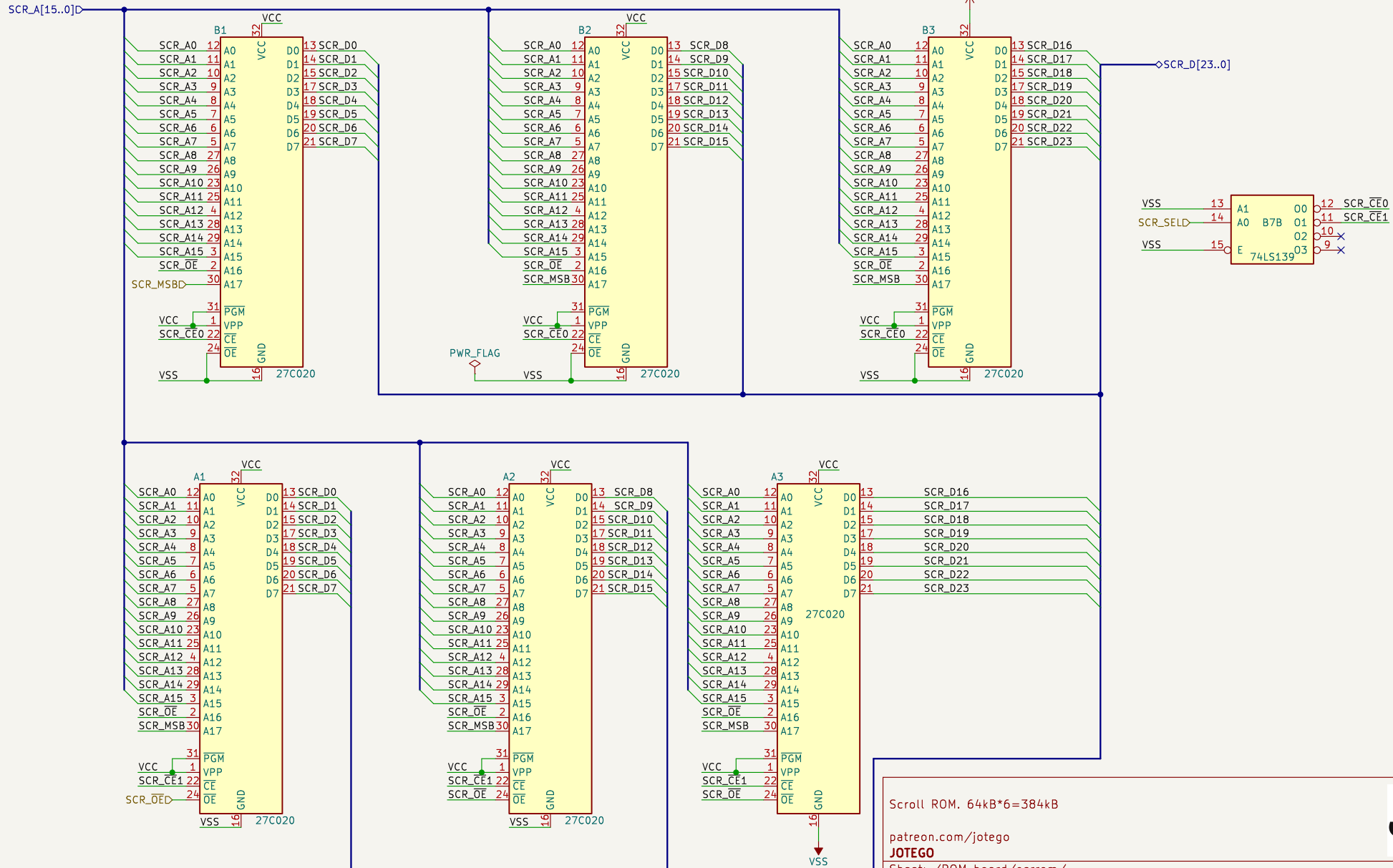
Title: SEGA System 18 MODEL 837-7178

Size: A4
KiCad E.D.A. 8.0.9

Date: 2022-03-18

Rev: José Tejada
Id: 5/26

SCROLL ROM



Scroll ROM. 64kB*6=384kB

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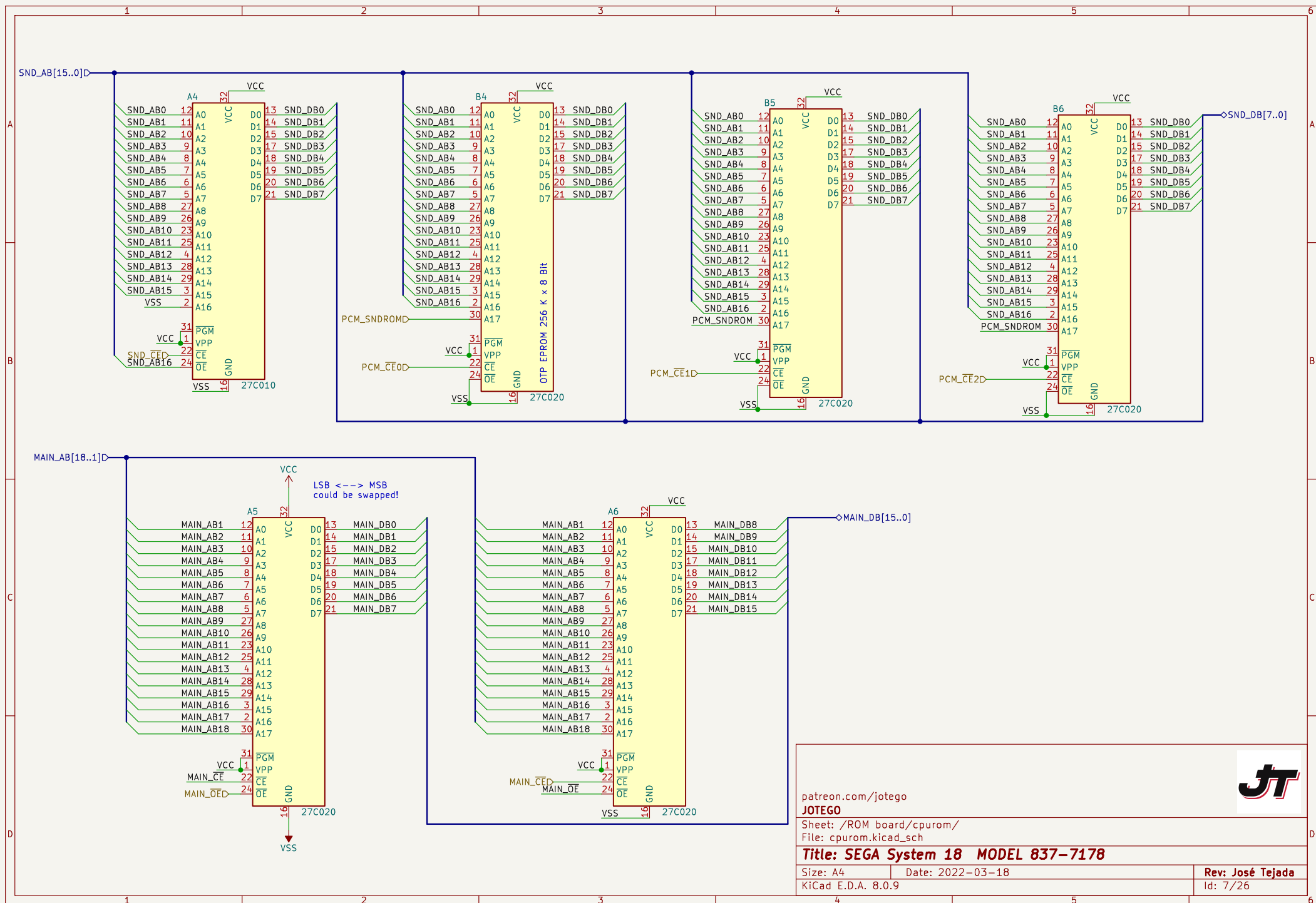
Sheet: /ROM board/scrrom/
File: scrrom.kicad_sch

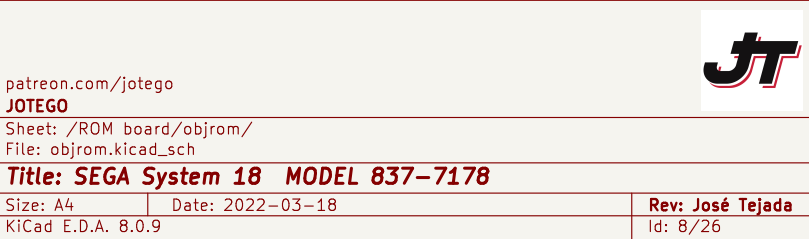
Title: SEGA System 18 MODEL 837-7178

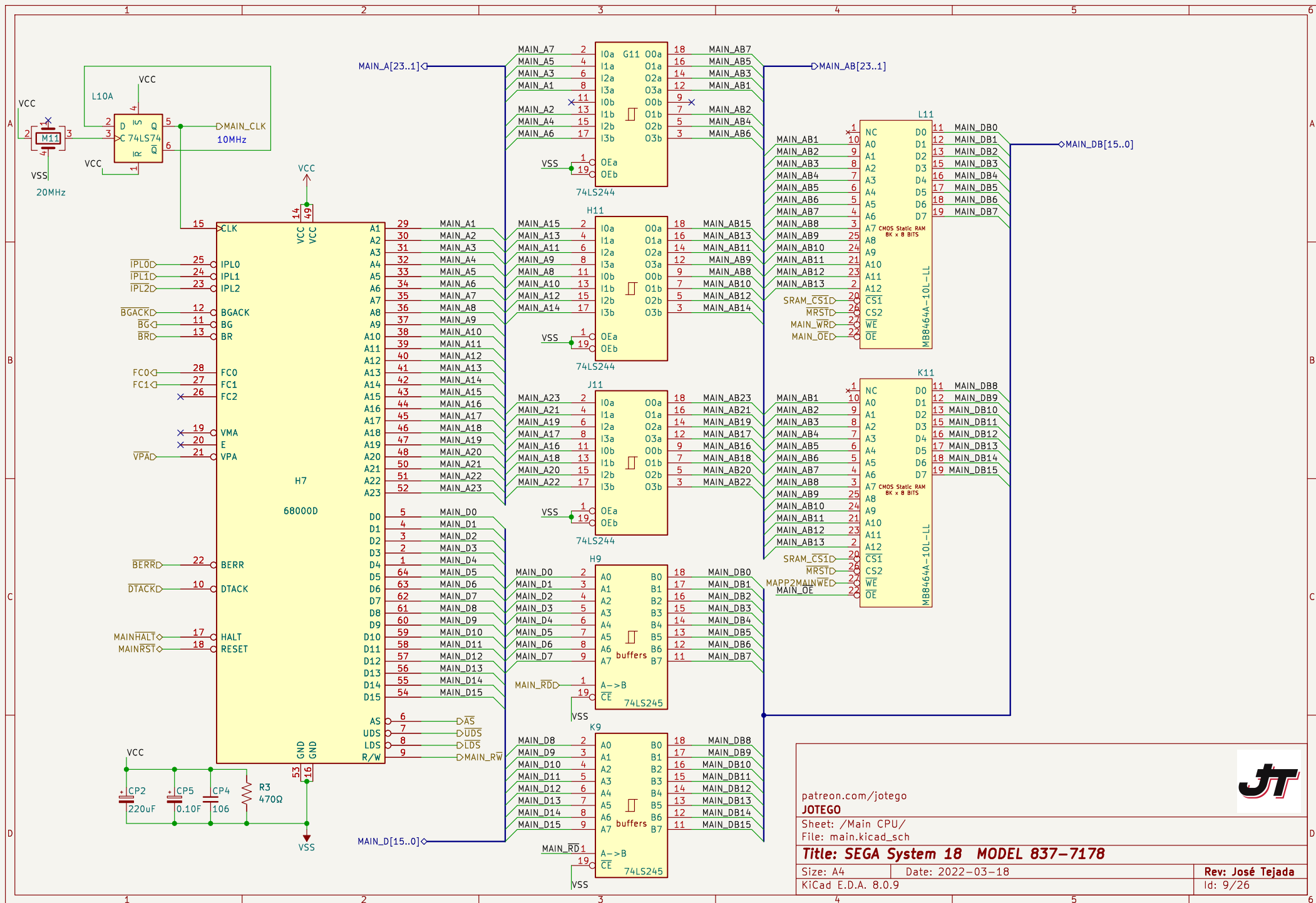
Size: A4 Date: 2022-03-18
KiCad E.D.A. 8.0.9

Rev: José Tejada
Id: 6/26









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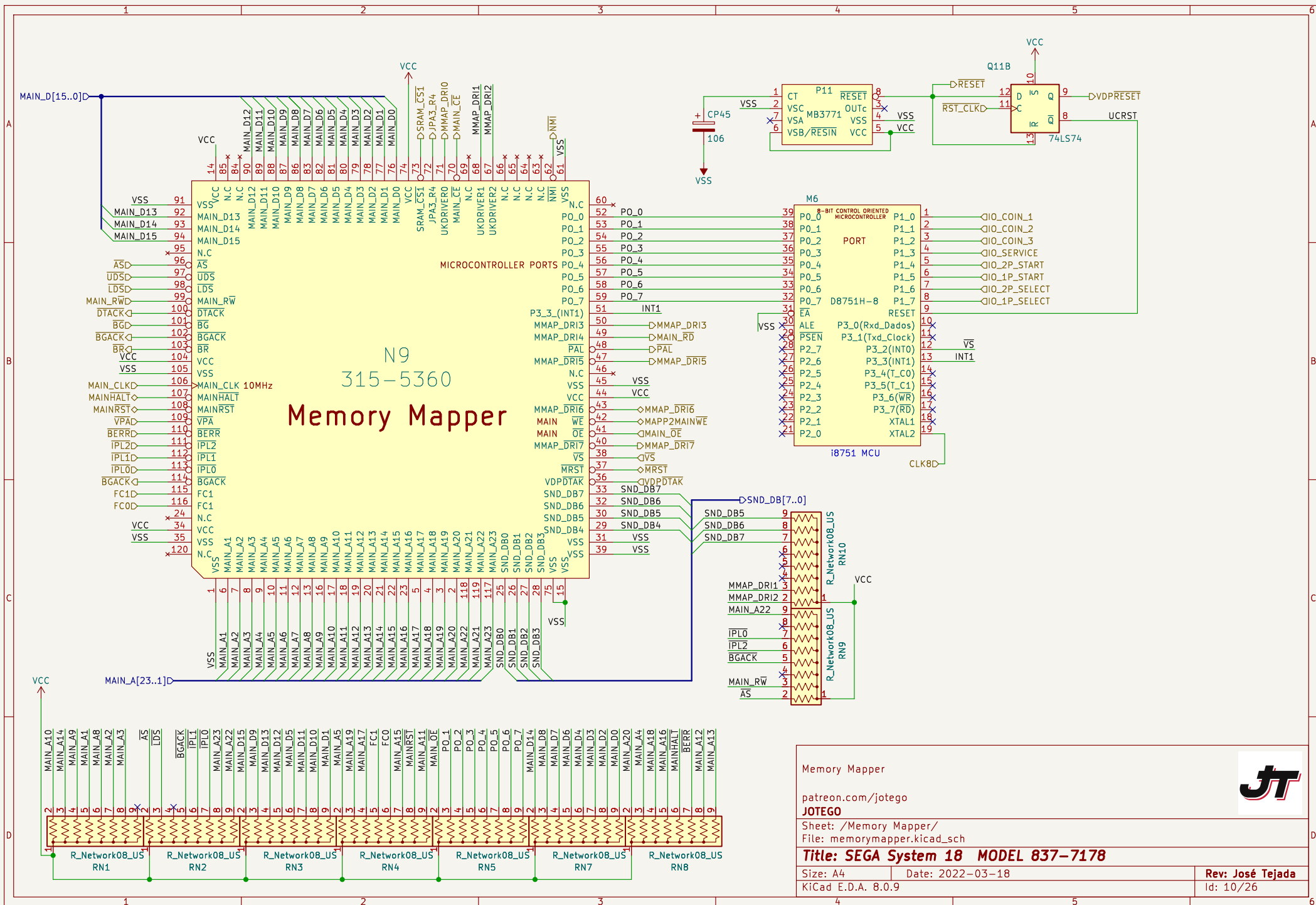
Sheet: /Main CPU/
File: main.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4 Date: 2022-03-18
KiCad E.D.A. 8.0.9

Rev: José Tejada
Id: 9/26





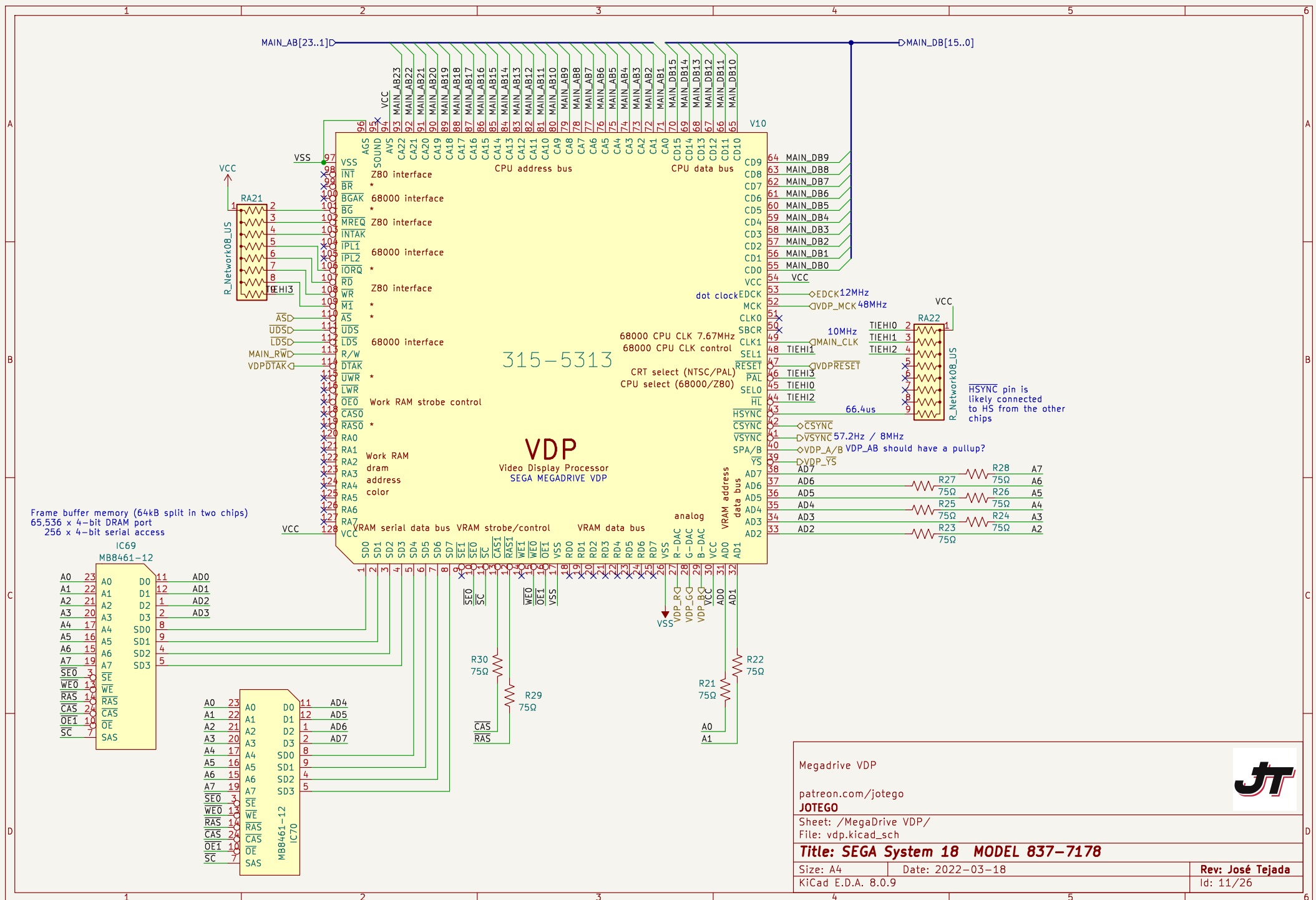
Memory Mapper

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JOTEGO

Sheet: /Memory Mapper/
File: memorymapper.kicad_sch

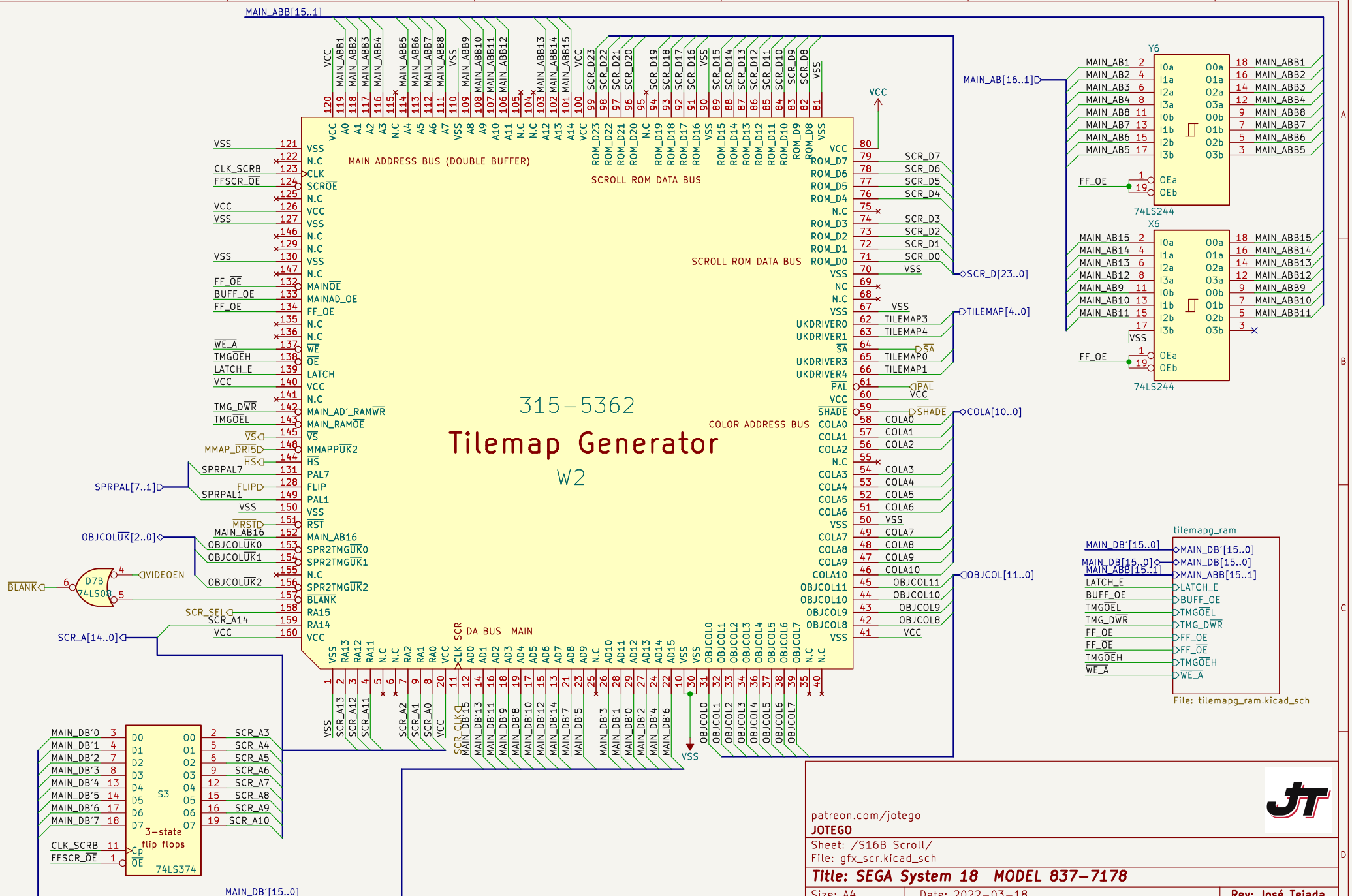
Title: SEGA System 18 MODEL 837-7178

Size: A4 Date: 2022-03-18
KiCad E.D.A. 8.0.9 Rev: José Tejada
Id: 10/26



Tilemap Generator

315-5362
W2



File: tilemapg_ram.kicad_sch



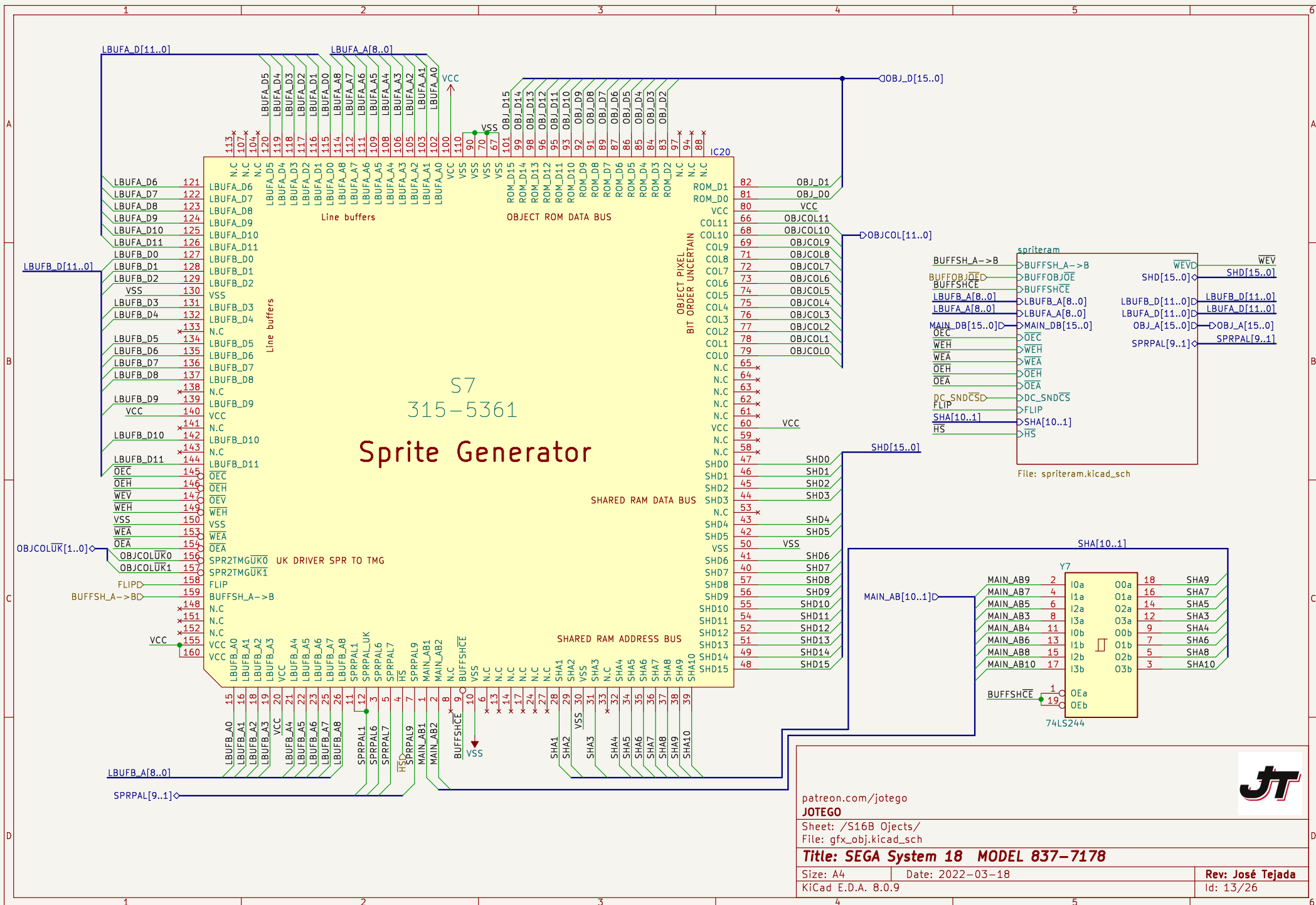
patreon.com/jotego
JOTEGO

Sheet: /S16B Scroll/
File: gfx_scr.kicad_sch

Title: SEGA System 18 MODEL 837-7178

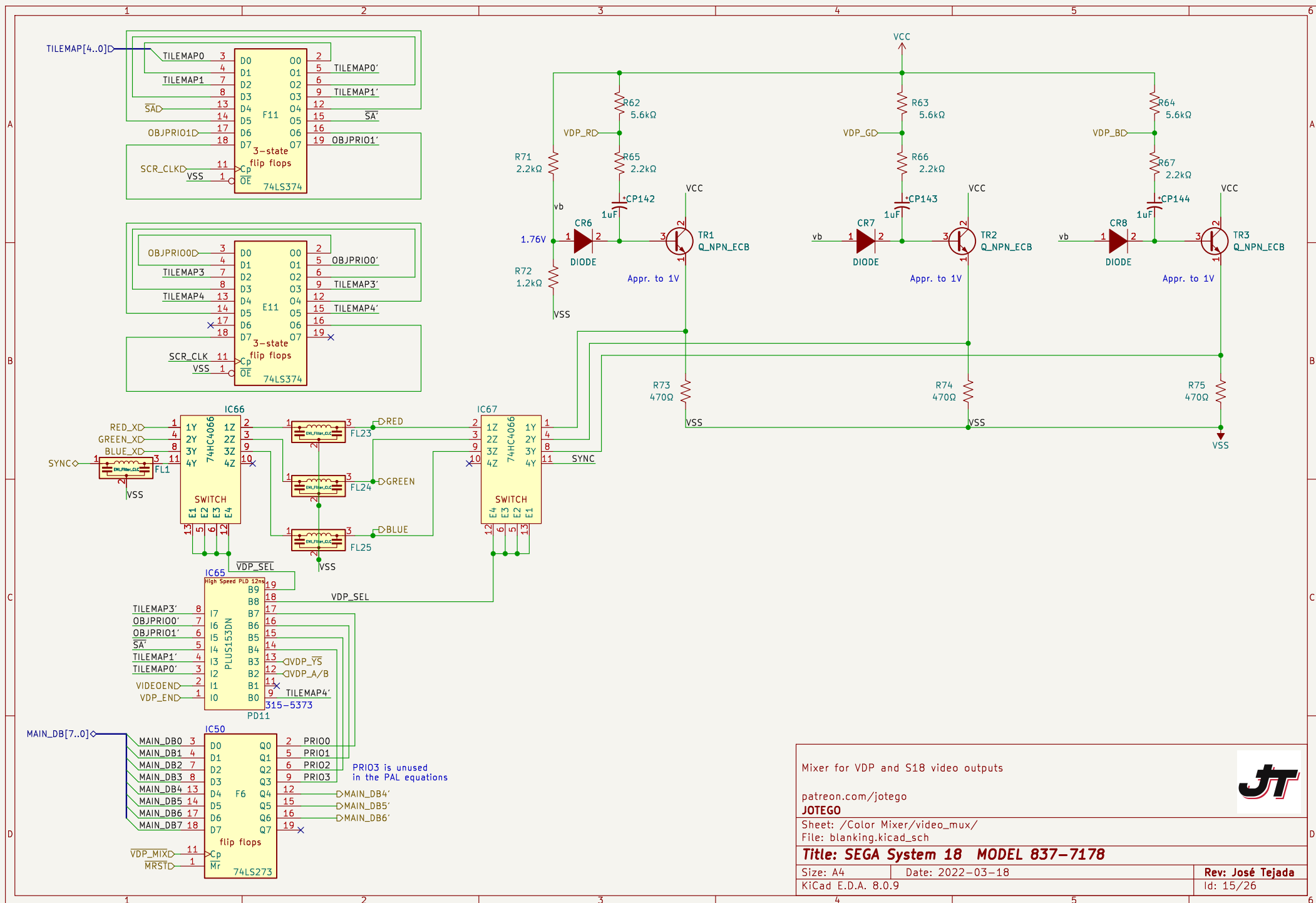
Size: A4
Date: 2022-03-18
KiCad E.D.A. 8.0.9

Rev: José Tejada
Id: 12/26





Rev: José Tejada
Id: 14/26



Mixer for VDP and S18 video outputs

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Sheet: /Color Mixer/video_mux/

File: blanking.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4 Date: 2022-03-18

KiCad E.D.A. 8.0.9

Rev: José Tejada

Id: 15/26





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JOTEGO

Sheet: /Capacitors/
 File: capacitors.kicad_sch

Title: SEGA System 18 MODEL 837-7178

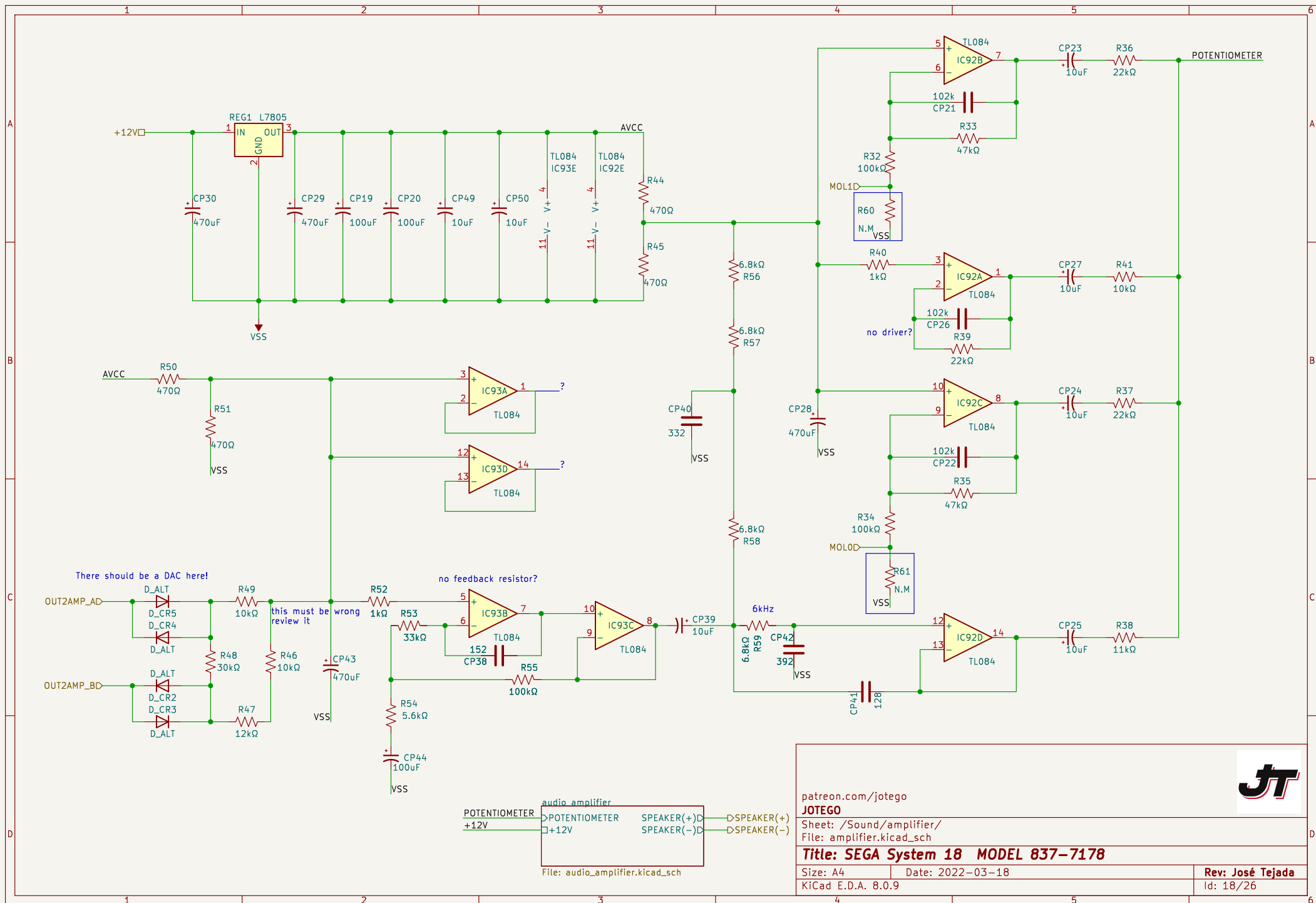
Size: A4 Date: 2022-03-18
 KiCad E.D.A. 8.0.9

Rev: José Tejada
 Id: 16/26





Rev: José Tejada
Id: 17/26



MOTHER AND DAUGHTER PCB CONNECTORS

Connector CN1:

Pin	Signal	Pin	Signal
a1	SCR_D0	b1	SCR_D1
a2	SCR_D2	b2	SCR_D3
a3	SCR_D4	b3	SCR_D5
a4	SCR_D6	b4	SCR_D7
a5	SCR_D8	b5	SCR_D9
a6	SCR_D10	b6	SCR_D11
a7	SCR_D12	b7	SCR_D13
a8	SCR_D14	b8	SCR_D15
a9	SCR_D16	b9	SCR_D17
a10	SCR_D18	b10	SCR_D19
a11	SCR_D20	b11	SCR_D21
a12	SCR_D22	b12	SCR_D23
a13	SCR_A13	b13	SCR_A14
a14	SCR_A11	b14	SCR_A12
a15	SCR_A9	b15	SCR_A10
a16	SCR_A7	b16	SCR_A8
a17	SCR_A5	b17	SCR_A6
a18	SCR_A3	b18	SCR_A4
a19	SCR_A1	b19	SCR_A2
a20	SCR_A15	b20	SCR_A0
a21	JPB3_R0	b21	JPB3_R1
a22	JPB3_R2	b22	JPB3_R1
a23	VDPDTAK	b23	MAINBUFF_CLK
a24	PRE_SYNCD	b24	
a25	VSS	b25	VSS
a26	VSS	b26	VSS
a27	VSS	b27	VSS
a28	VSS	b28	VSS
a29	VCC	b29	VCC
a30	VCC	b30	VCC

Connector CN2:

Pin	Signal	Pin	Signal
a1	OBJ_D0	b1	OBJ_D1
a2	OBJ_D2	b2	OBJ_D3
a3	OBJ_D4	b3	OBJ_D5
a4	OBJ_D6	b4	OBJ_D7
a5	OBJ_D8	b5	OBJ_D9
a6	OBJ_D10	b6	OBJ_D11
a7	OBJ_D12	b7	OBJ_D13
a8	OBJ_D14	b8	OBJ_D15
a9	OBJ_A19	b9	OBJ_A16
a10	OBJ_A18	b10	OBJ_A17
a11	OBJ_A0	b11	OBJ_A1
a12	OBJ_A2	b12	OBJ_A3
a13	OBJ_A4	b13	OBJ_A5
a14	OBJ_A6	b14	OBJ_A7
a15	OBJ_A8	b15	OBJ_A9
a16	OBJ_A10	b16	OBJ_A11
a17	OBJ_A12	b17	OBJ_A13
a18	OBJ_A14	b18	OBJ_A15
a19	OBJ_A16	b19	OBJ_A17
a20	OBJ_A18	b20	OBJ_A19
a21	OBJ_A20	b21	OBJ_A21
a22	OBJ_A22	b22	OBJ_A23
a23	OBJ_A24	b23	OBJ_A25
a24	OBJ_A26	b24	OBJ_A27
a25	OBJ_A28	b25	OBJ_A29
a26	OBJ_A30	b26	OBJ_A31
a27	OBJ_A32	b27	OBJ_A33
a28	OBJ_A34	b28	OBJ_A35
a29	OBJ_A36	b29	OBJ_A37
a30	OBJ_A38	b30	OBJ_A39
a31	OBJ_A40	b31	OBJ_A41
a32	OBJ_A42	b32	OBJ_A43
a33	OBJ_A44	b33	OBJ_A45
a34	OBJ_A46	b34	OBJ_A47
a35	OBJ_A48	b35	OBJ_A49
a36	OBJ_A50	b36	OBJ_A51
a37	OBJ_A52	b37	OBJ_A53
a38	OBJ_A54	b38	OBJ_A55
a39	OBJ_A56	b39	OBJ_A57
a40	OBJ_A58	b40	OBJ_A59
a41	OBJ_A60	b41	OBJ_A61
a42	OBJ_A62	b42	OBJ_A63
a43	OBJ_A64	b43	OBJ_A65
a44	OBJ_A66	b44	OBJ_A67
a45	OBJ_A68	b45	OBJ_A69
a46	OBJ_A70	b46	OBJ_A71
a47	OBJ_A72	b47	OBJ_A73
a48	OBJ_A74	b48	OBJ_A75
a49	OBJ_A76	b49	OBJ_A77
a50	OBJ_A78	b50	OBJ_A79
a51	OBJ_A80	b51	OBJ_A81
a52	OBJ_A82	b52	OBJ_A83
a53	OBJ_A84	b53	OBJ_A85
a54	OBJ_A86	b54	OBJ_A87
a55	OBJ_A88	b55	OBJ_A89
a56	OBJ_A90	b56	OBJ_A91
a57	OBJ_A92	b57	OBJ_A93
a58	OBJ_A94	b58	OBJ_A95
a59	OBJ_A96	b59	OBJ_A97
a60	OBJ_A98	b60	OBJ_A99
a61	OBJ_A100	b61	OBJ_A101
a62	OBJ_A102	b62	OBJ_A103
a63	OBJ_A104	b63	OBJ_A105
a64	OBJ_A106	b64	OBJ_A107
a65	OBJ_A108	b65	OBJ_A109
a66	OBJ_A110	b66	OBJ_A111
a67	OBJ_A112	b67	OBJ_A113
a68	OBJ_A114	b68	OBJ_A115
a69	OBJ_A116	b69	OBJ_A117
a70	OBJ_A118	b70	

Used as ADC
inputs from gun
in Laser Ghost
on board:
837-7536
171-5993A

JOTEGO

File: top_connectors.kicad_sch

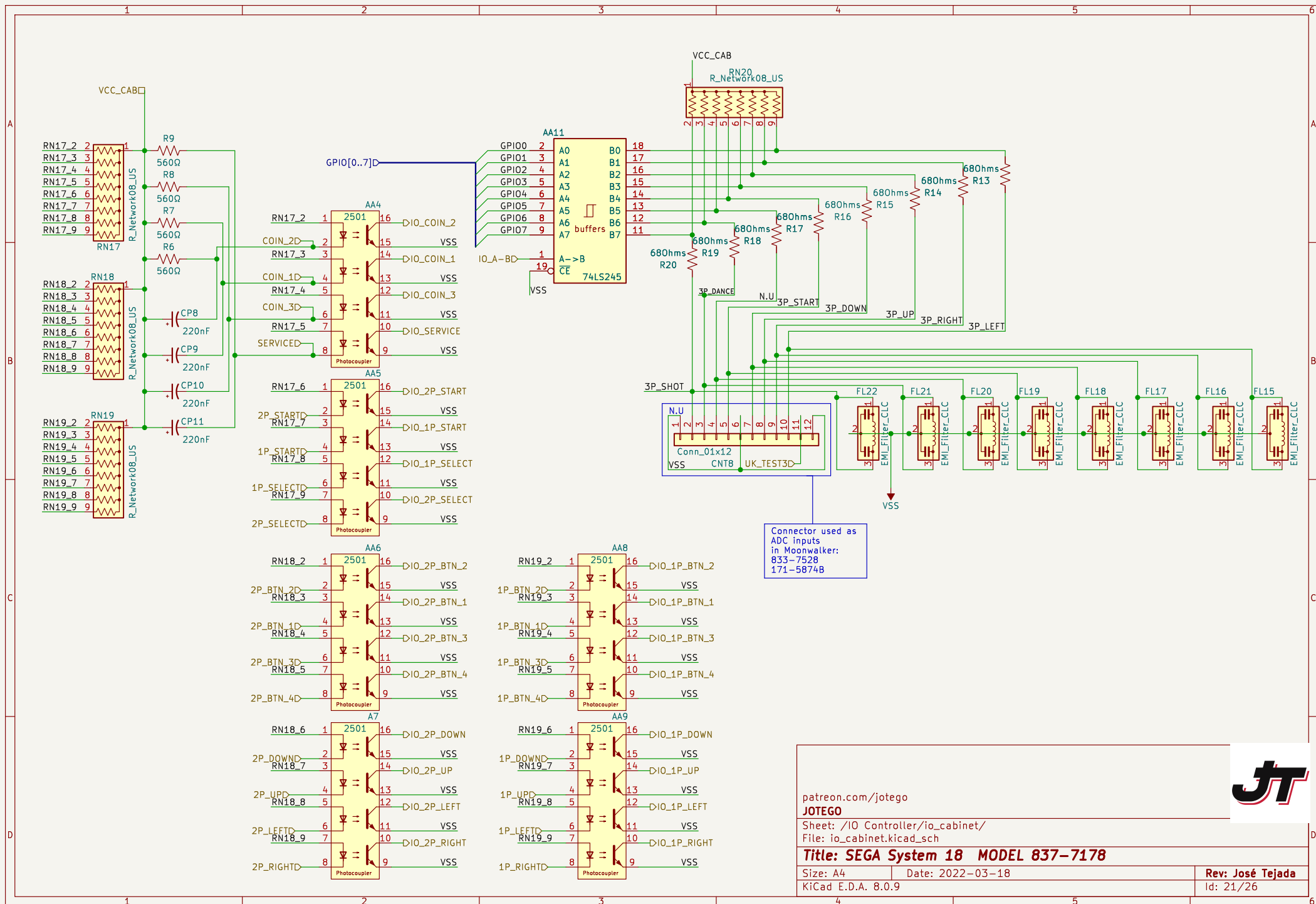
Title: SEGA System 18 MODEL 837-7178

Size: A4
KiCad E.D.A. 8.0.9

Id: 19/26







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Sheet: /IO Controller/io_cabinet/
File: io_cabinet.kicad_sch

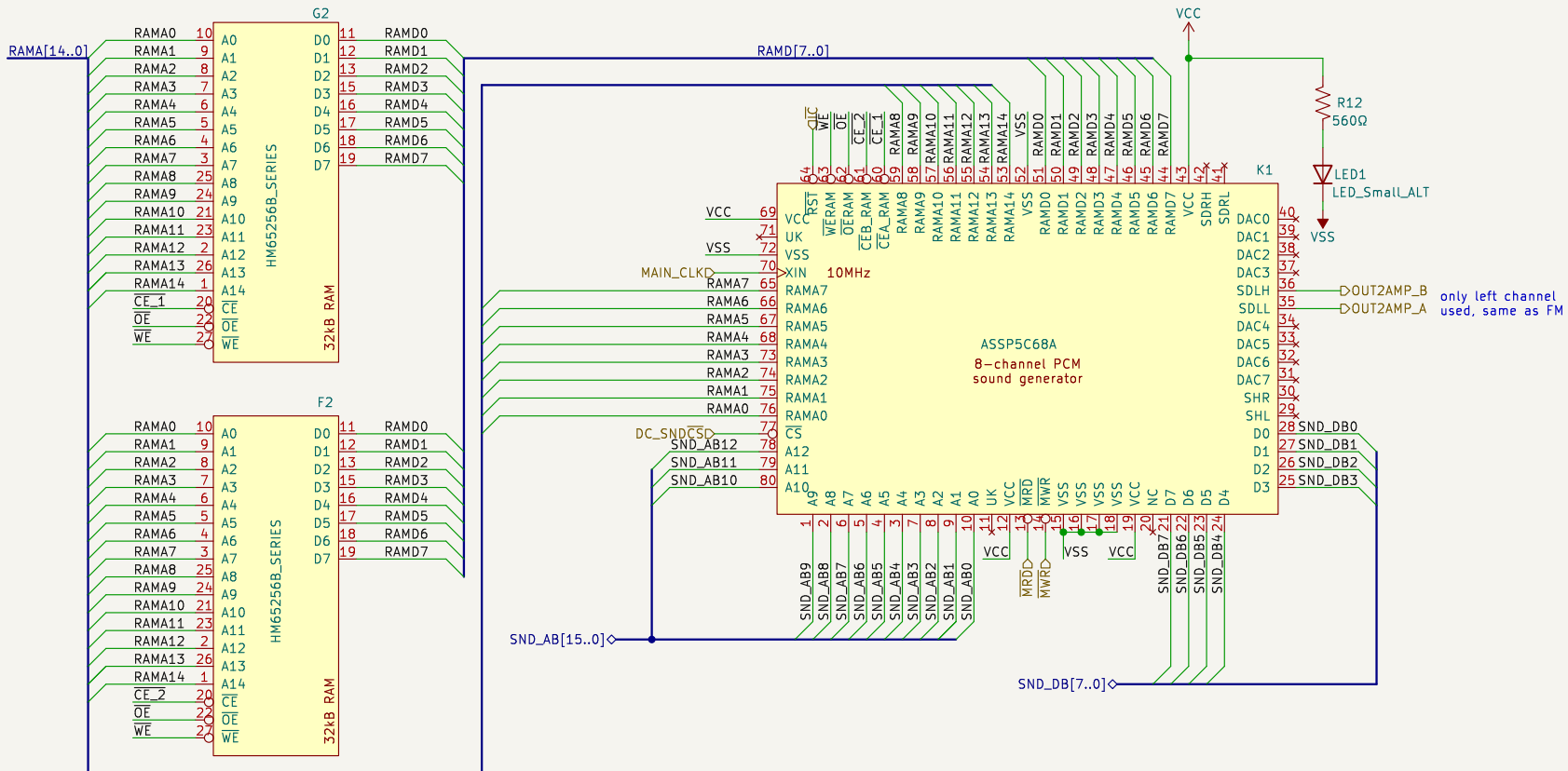
Title: SEGA System 18 MODEL 837-7178

Size: A4
KiCad E.D.A. 8.0.9

Date: 2022-03-18

Rev: José Tejada
Id: 21/26





PCM sound generator

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Sheet: /Sound/PCM/

File: comm_snd.kicad_sch

Title: SEGA System 18 MODEL 837-7178

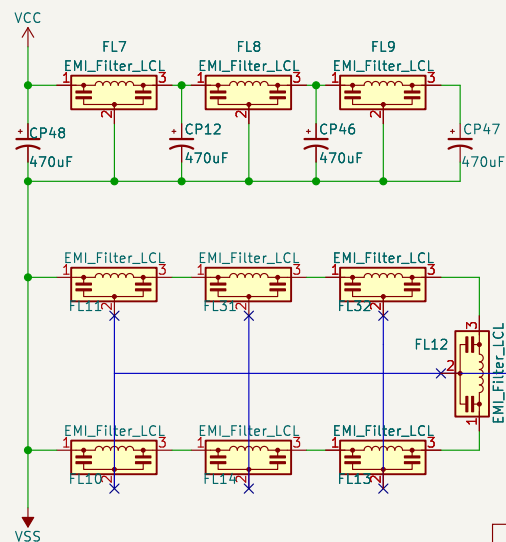
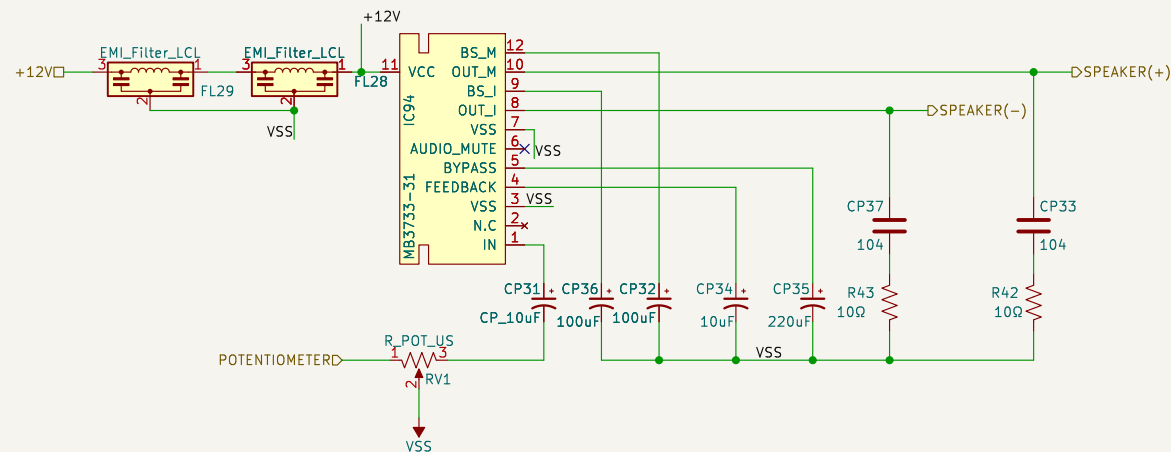
Size: A4 Date: 2022-03-18

KiCad E.D.A. 8.0.9

Rev: José Tejada

Id: 22/26





Leaded EMIFL
Three-terminal Capacitor type
DSS6NE52A222p
2200pF 80/-20%
100Vdc 6A -25 to +85°C

A kind of a track is observed between the two PCB layers, as if it were another layer, but there is no certainty about that. The track seems to be used to delimit or isolate the EMI (ElectroMagnetic Interference) filters. They may have wanted to isolate the noise between digital and analog devices.

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JOTEGO

Sheet: /Sound/amplifier/audio_amplifier/
File: audio_amplifier.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4

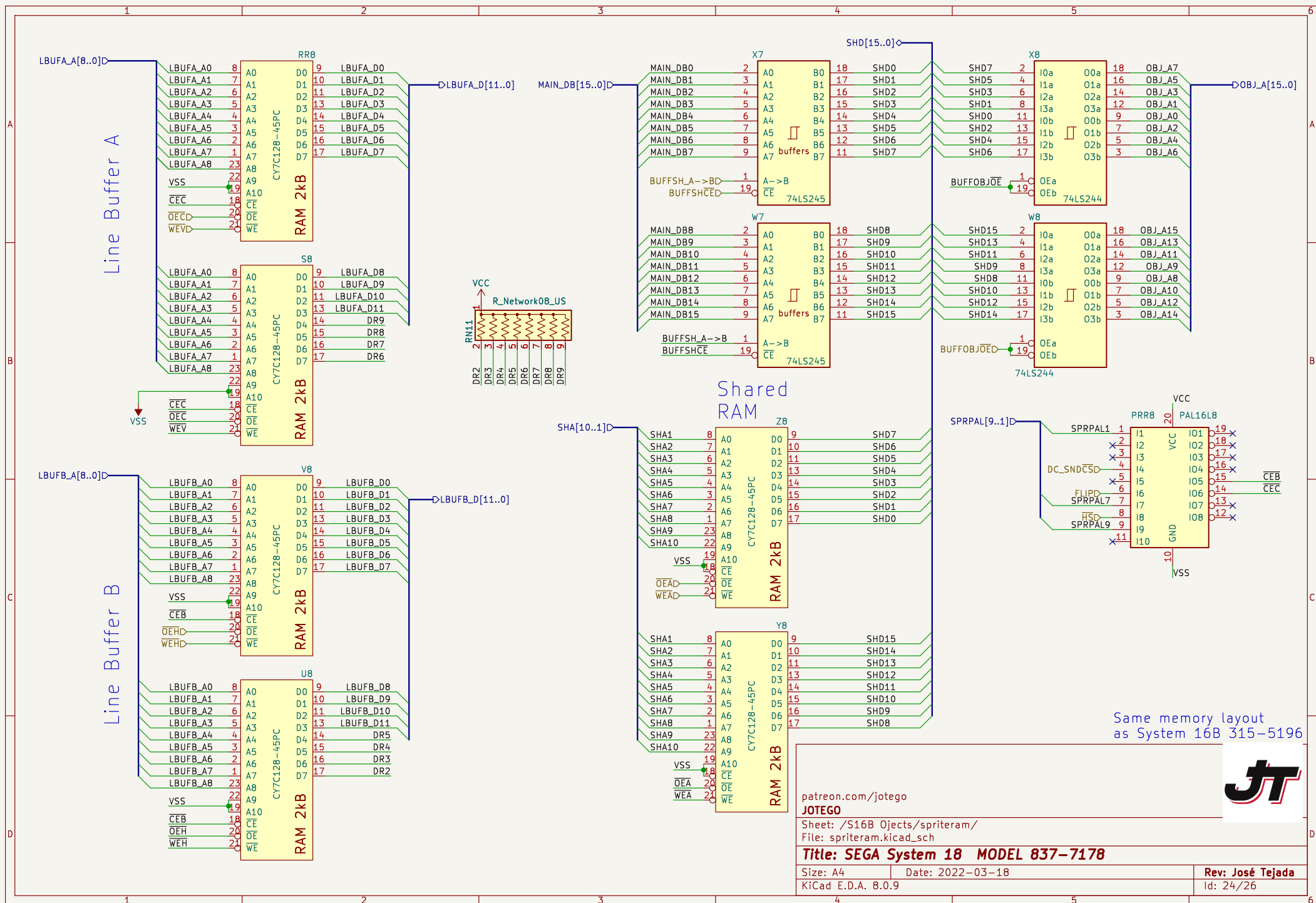
Date: 2022-03-18

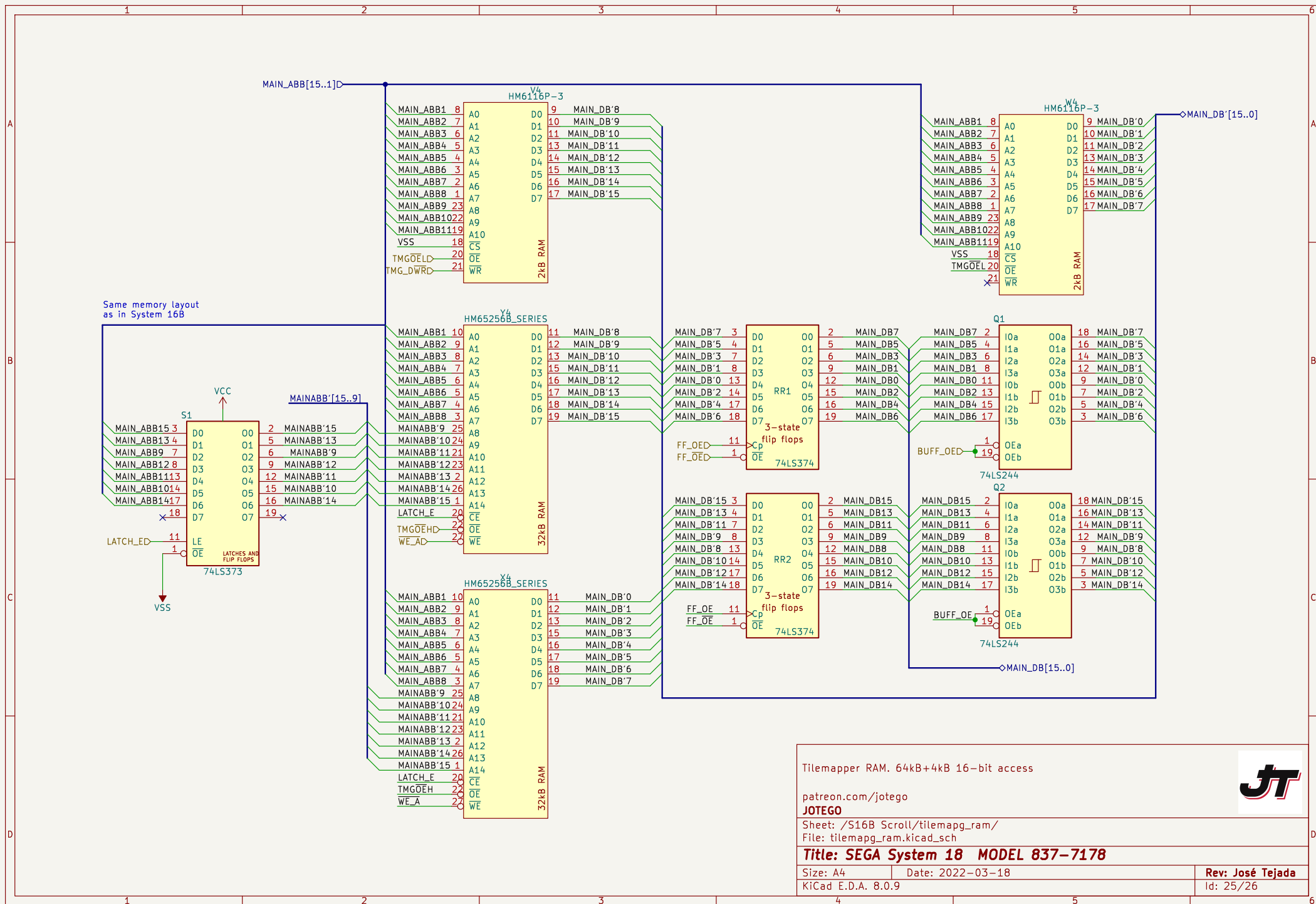
KiCad E.D.A. 8.0.9

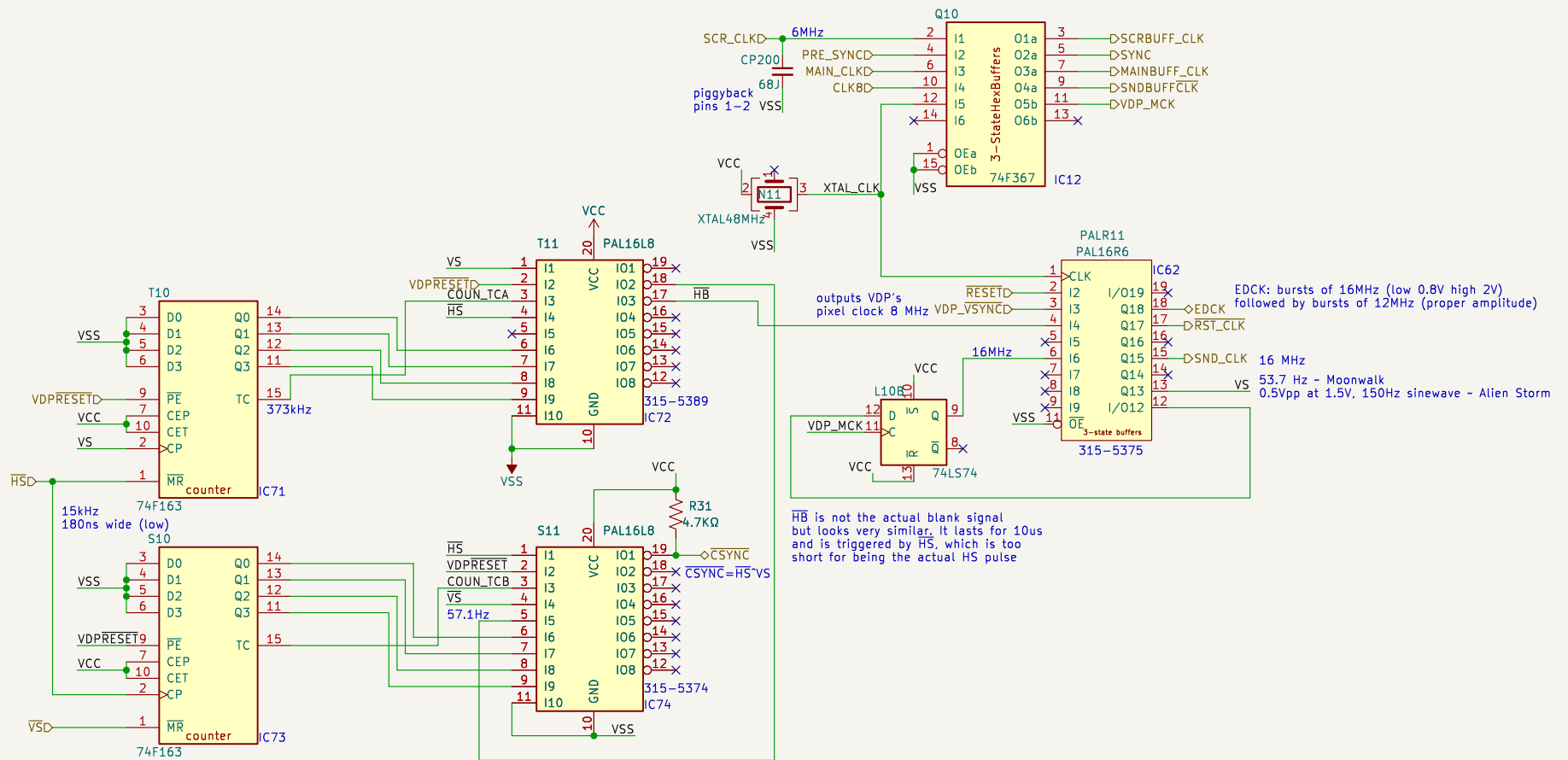
Rev: José Tejada

Id: 23/26









VDP pixel clock and csync generation
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Sheet: /Video Sync/
File: buffers.kicad_sch

Title: SEGA System 18 MODEL 837-7178

Size: A4 Date: 2022-03-18

KiCad E.D.A. 8.0.9

Rev: José Tejada

Id: 26/26

